

# Unfolding the Leech Lattice: Fused Multi-Shell Decoding and VRAM Layouts for 2-Bit LLM Weights

What a combinatorial index costs to serve, measured against deployed 4-bit and 2-bit GEMV kernels

## Author

Pier-Jean Malandrino  
Scub, Bordeaux, France  
[pierjean.malandrino@scub.net](mailto:pierjean.malandrino@scub.net)

Corresponding author, and sole author.

## Manuscript type

New Paper, Not an Extension of a Conference Paper. The work has not been published before in any form and is not under review elsewhere.

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## Artifact availability

The manuscript's availability section withholds these URLs for double-anonymous review. They are, for the editor:

Code, data and paper source: <https://github.com/pjmalandrino/llvq>

Quantized 4B artifact: <https://huggingface.co/Pier-Jean/Qwen3-4B-LLVQ-2bit>

The repository is public. The manuscript itself is fully anonymized and carries no link that identifies the author; the URLs are given here, for the editor, so that the artifact can be located without the manuscript breaking its own anonymity.

## Use of generative AI

Disclosed under ACM's Policy on Authorship, and repeated in the manuscript itself. Anthropic's Claude, through the Claude Code CLI in 2026, drafted and revised the manuscript's text, wrote the figure-generating and table-checking scripts, wrote portions of the CUDA kernels, Rust host code and tests, and recomputed statistics from the committed measurement journals. The author designed the study, authorized every measurement, verified each number against its source, and is responsible for the entire content.